Experiment 5 – Transistor Radio

Justin Lewis

Physics 4700 – Dr. Tom Barrett

Abstract:

In this experiment, we extended the early amplifier stages of the previous experiment into a multistage transistor AM radio system. The fourth stage transistor was built as a common emitter and measured at 1Mhz so we could acquire the operating point and AC voltage gain. Also, an input transformer alongside a voltage divider we placed at the front end of the circuit to allow a small signal for larger gain operations. An L-C tuning circuit was placed in between stages 2 and 3 where the resonate response was measured so that the quality factor Q could be found. Later, a rectifier stage was added to recover any audio frequency envelope from the AM carrier, thus a low-pass capacitor was also added to observer and listen to the extracted audio signal. Load line analysis was to be performed to determine why the recitifier stages behavior is asymmetric while stage 4 doesn't behave this way. Finally, the final stage implemented was the output amplifier stage with a range over 100Hz to 20KHz, and to complete the circuit, an antenna was added to receive local AM radio stations. The experiment verifys how multiple transistor amplifier stages alongside resonance, rectification, and pass filtering put together can be used as a radio receiver.

1: Stage 4 Common-Emitter

Here the common-emitter stage was designed to have a target gain around 20 with a collector current of 10 mA. According to the design requirements, this stage needed a 470 Ω collector resistor that was to be in parallel with a 50-pF capacitor, and a 10 nF input coupling capacitor. The operating points at this stage include the following:

Collector = 5.31V, Base = 0.921V, Emitter = 0.206V

These operating points show that the transistor was biased in its region, and the difference between the base and the emitter was close to the expected junction drop of about 0.7 V, so the bias was consistent with the proper transistor operation. The collector current can be estimated from the voltage drop across the 470 Ω collector resistor, which was calculated to be 9.98 mA, which is very close to the expected 10 mA.

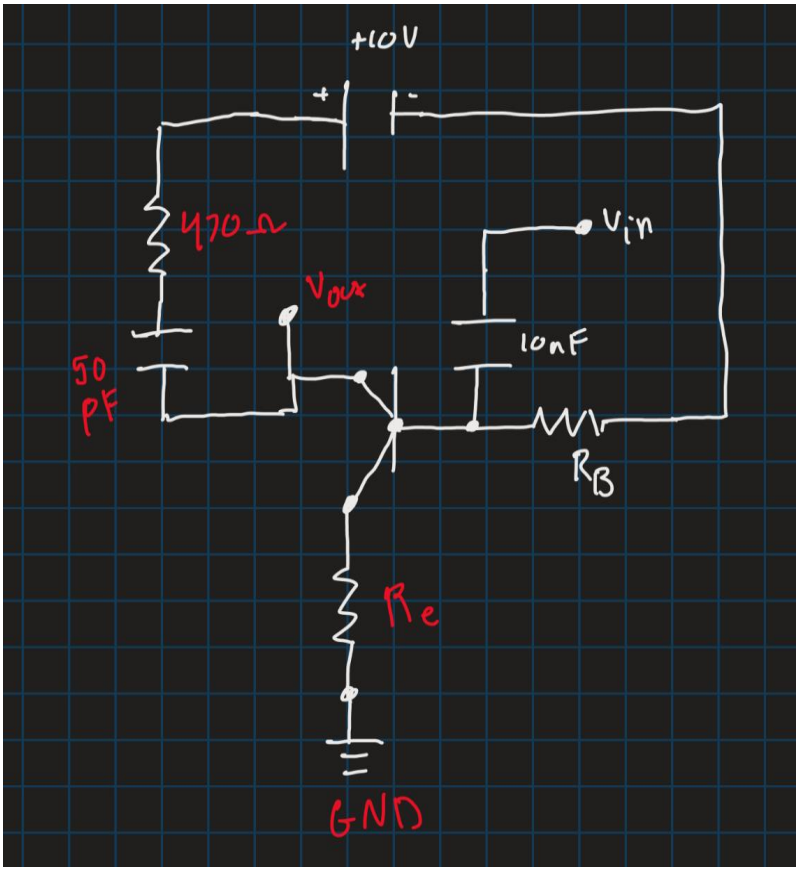


Figure 1.

2A: Input Transforer and Voltage Divider

The input transformer was added in front of stage 1, and could be considered stage 0. The gain of the transistor chain had become too large to allow the function generator to directly give a small enough signal, we then added a voltage divider before the transformer in hopes of reducing the signal amplitude going into the circuit. A voltage divider of 1kΩ and 10 Ω was used to get a ratio of 1/101 with a measured FG output of 36.5mV, and thus the input voltage into the transformer was calculated to be about 0.361mV. This allowed the to be driven at a much smaller signal level to prevent overdriving the high gain amplifier stages. The mini-transformers operation was verified by the oscilloscope as seen in the figure below.

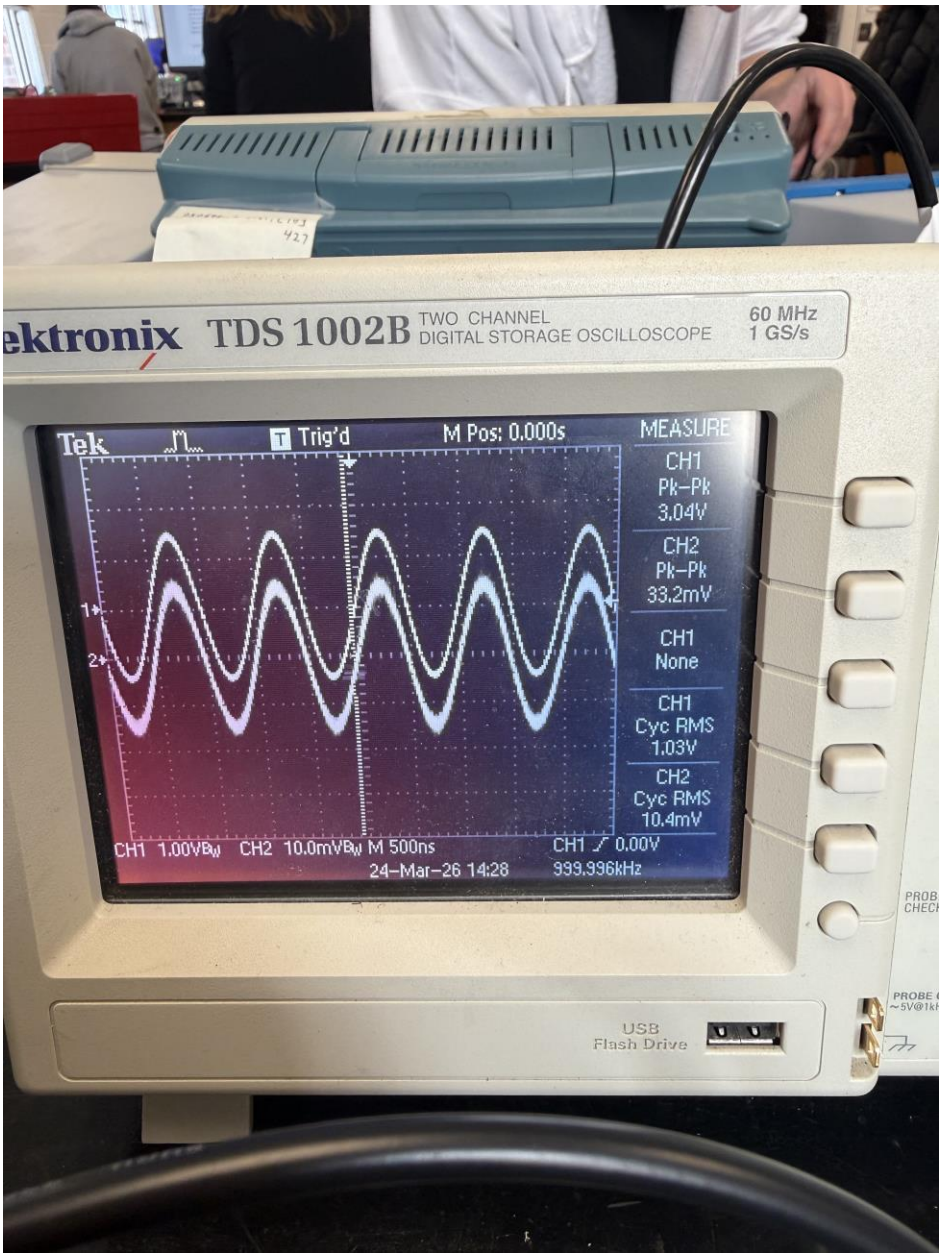


Figure 2.

2B:

To test the grounding effectiveness for stage 3, we applied a 1 MHz signal to the mini-transformer and measured the output of stage 4. There were two measurements taken between stages 2 and 3, one taken with the coupling capacitor, and the other taken without the coupling capacitor in between.

With the coupling capacitor in place:

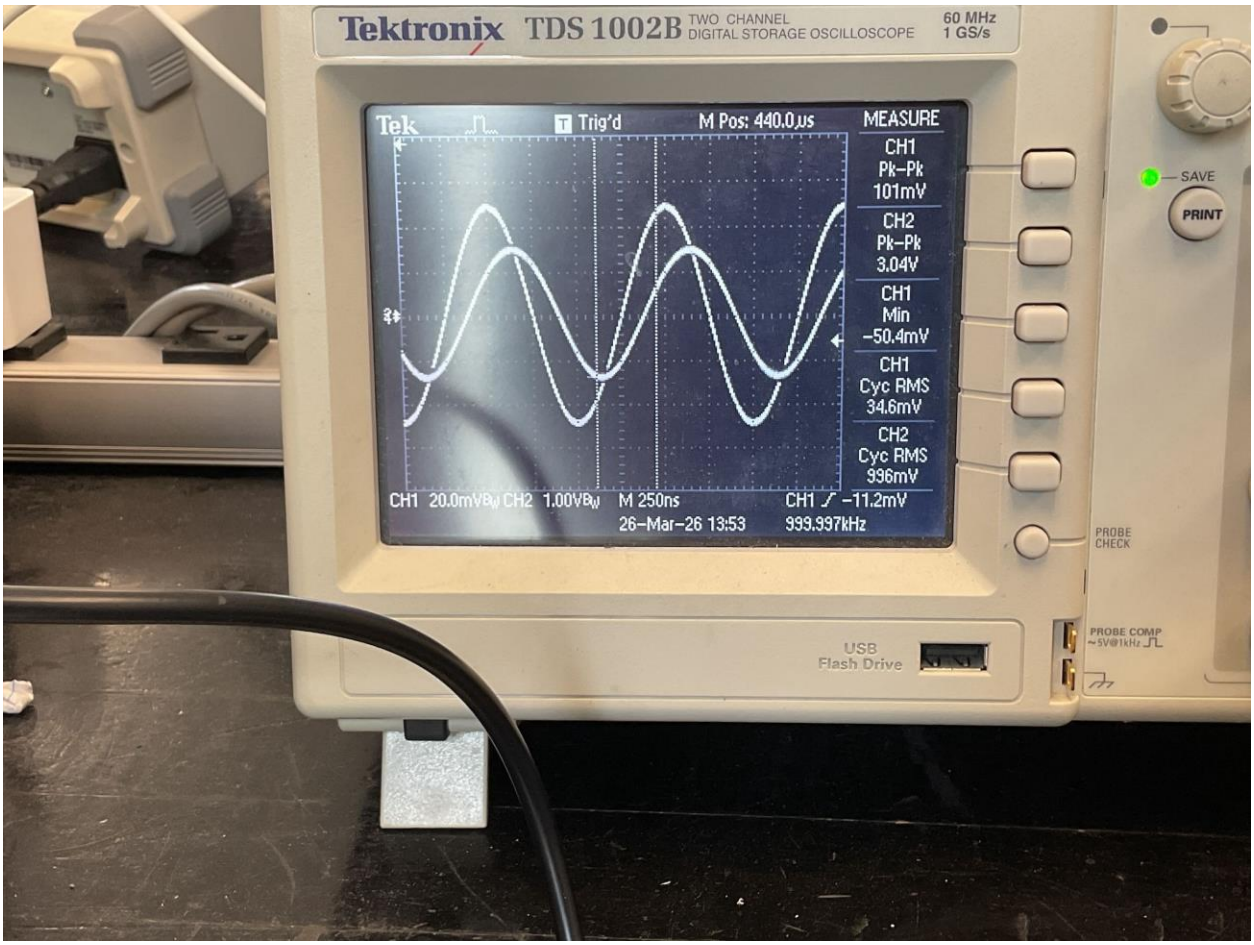


Figure 3.

Without the coupling capacitor in place:

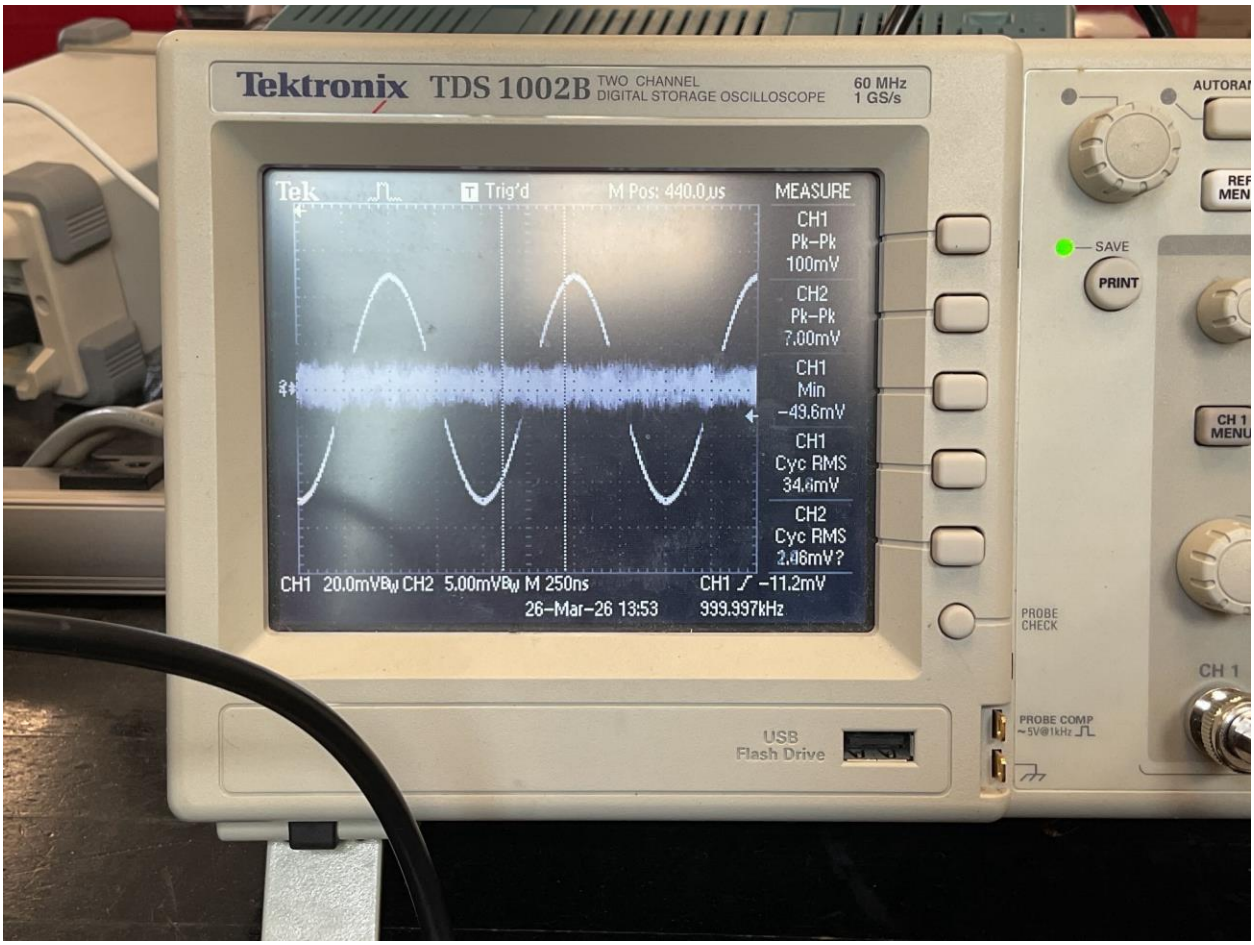


Figure 4.

3: Tuning Circuit and Measurement of Q

The resonant frequency was measured after the L-C tuning circuit was inserted betwen the emitter resistor of stage 2 and the emitter input of stage 3 in place of the coupling capacitor. A 1 MHz input signal was applied to the mini-transformer and the tuning capacitor was adjusted to get a maximum output signal at stage 4. Once we found the strongest response, the frequnecy was varied to around the resonance to determine the resonant frequency and the two 3dB points needed to determine the quality factor Q.

The measured resonant frequency was at the peak of 999 kHz, the upper 3 dB frequency was 1.002 MHz, and the lower was at 0.995 kHz. The peak output amplitude was 1.20 V

$\Delta f = 1002 - 995 = 7$ kHz

Thus the quality factor was determined from

$$Q = \frac{f_0}{\Delta f} = \frac{999}{7} \approx 142.7$$

These results show that the tuning circuit was operating correctly, and the tuning circuit produced a resonant response that was clearly visible around 1 MHz(approx 999kHz). Since the measured Q was higher than 100, the circuit met the expectations of the experiment, and demonstrated the effectiveness of the radio receiver.

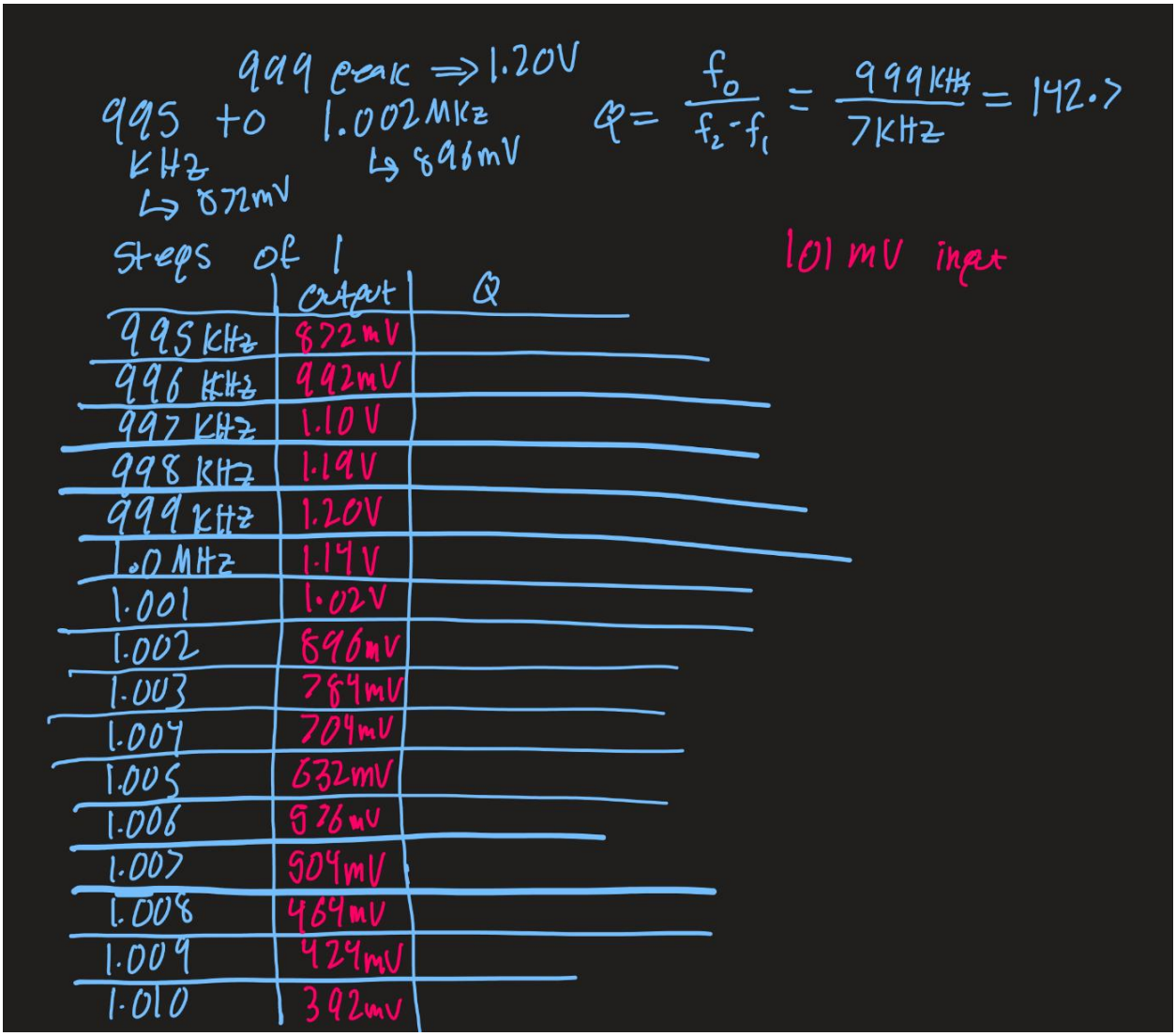


Figure 5.

4A: Rectifier Stage 5

Stage 5 was built to be a common-emitter rectifier stage for the radio circuit. Stage 5 utilized a 10 nF input coupling capacitor, a 470 Ω collector resistor, and a 220 Ω emitter resistor with. The bias adjusted so that the stage would provide an asymmetric gain that would produce the rectification of the incoming RF signal. Once rectificatiom was completed, a capacitor placed in parallel with the collector resistor would become a low-pass filter so the audio frequency could be extracted through the modulated carrier.

In the following figure, before the low-pass filtering was applied, the output waveform portrays a distorted and rectified high frequency signal near 1 MHz which is consistent with the behavior of the rectifier stage because the transistor was used to amplify one portion of the carrier more than the other.

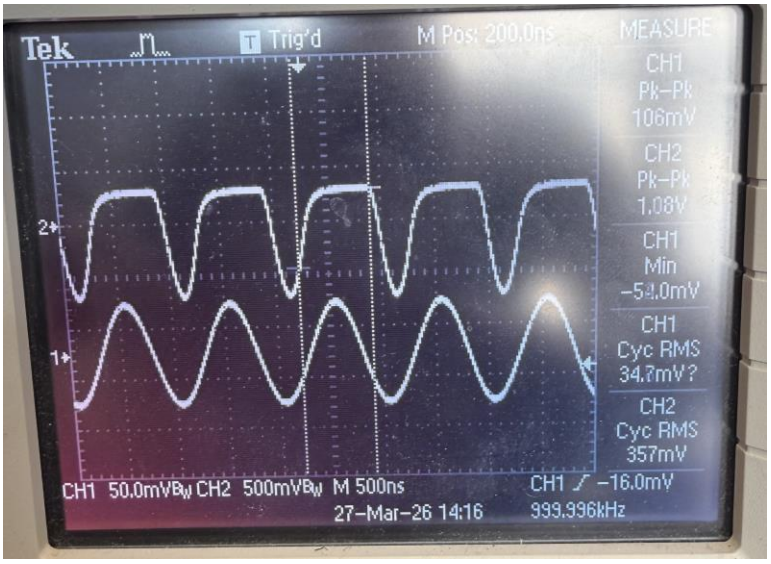


Figure 6.

Once the low-pass filtering was applied the source became modulated, and the output waveform provided the lower frequency recovered signal. The scope figure taken at 500 Hz shows that the audio envelope was in fact extracted from the modulated carrier. This was expected as the rectifier and low pass combination rectifies the carrier first, and the capacitor removes most of the RF component after, which leaves a slowly varying modulation signal.

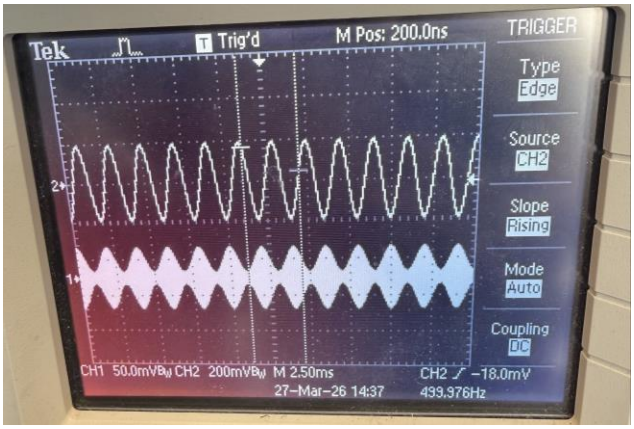


Figure 7.

4B: Load-Lines

5: Output Stage 6

Stage 6 was designed and built to be a common-emitter output stage. This stage was designed to use a 100 Ω collector resistor in parallel with a 220 nF capacitor, 2 µF input coupling capacitor, and a voltage gain of about 5 with a collector current of about 14 mA.

Using the design values and relations for this stage, the small-signal emitter resistance was estimated to be $r_e \approx \frac{25mV}{14mA} \approx 1.8\Omega$

And the target gain of about 5 gave an emitter resistance of 18.2 which was rounded to the nearest resistor value of 20 Ω, which gives an emitter voltage of 0.28 V and thus a base voltage of 0.98 V.

Assuming a beta of 150, the base current was estimated to be 14/150 which gives 0.0933 mA which then allows us to estimate the base resistor as 96.7 kΩ, where we used 100 kΩ. The collector voltage I failed to record, but the measured emitter and base voltages matched the design values well.

The AC gain of stage 6 was measured over the audio frequency range, which shows a gain a little over 4 which was reasonably close to the expected gain of 5. As the frequency increased, the gain decreased in a gradual manner which showcases the low-pass behavior of the capacitor in parallel with the collector resistor.

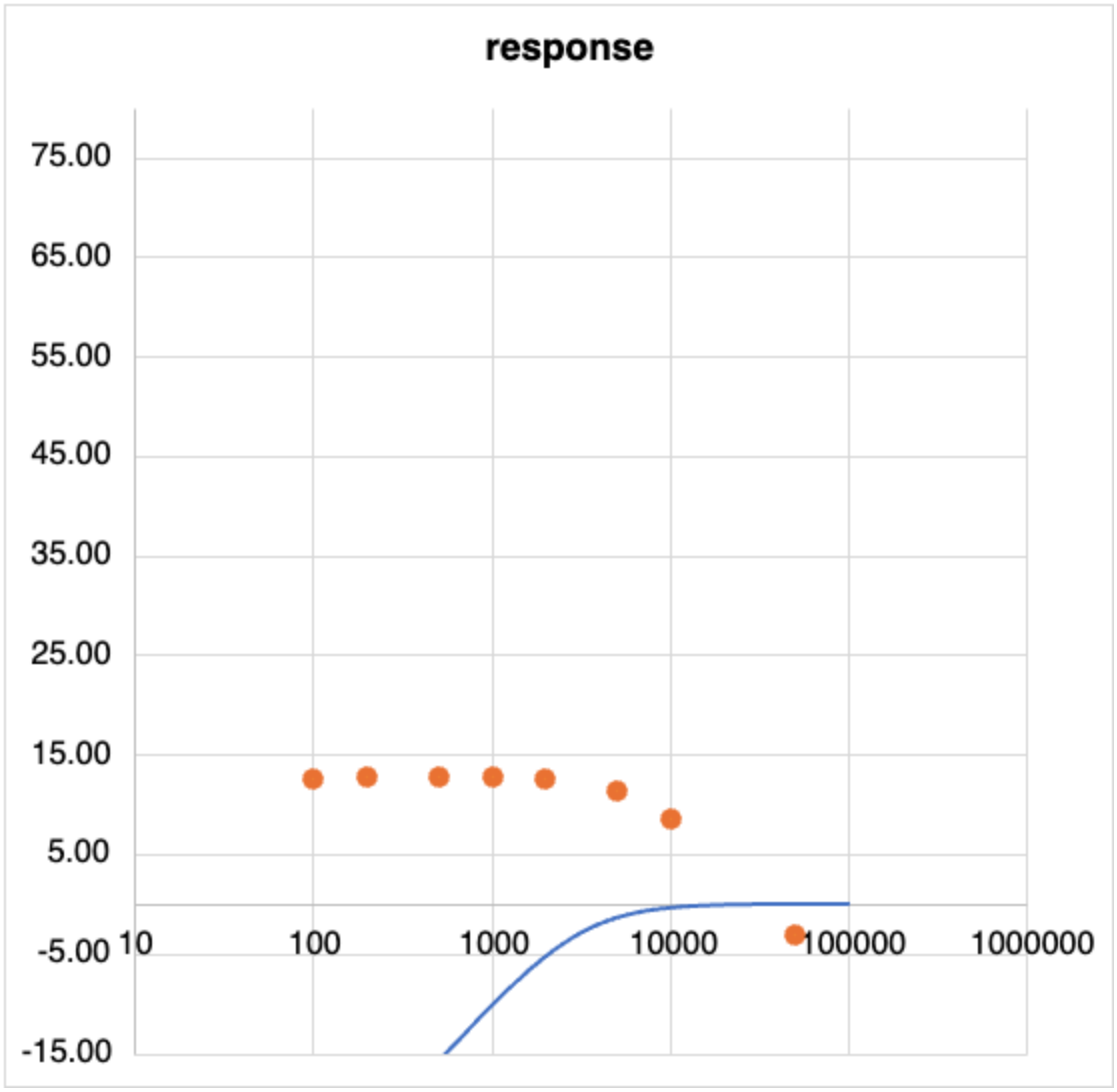


Figure 8.

6: Audio Channels

The following channel signals were tuned into:

1. 820 AM
2. 610 AM
3. 1460 AM

7: Total Schematic diagram

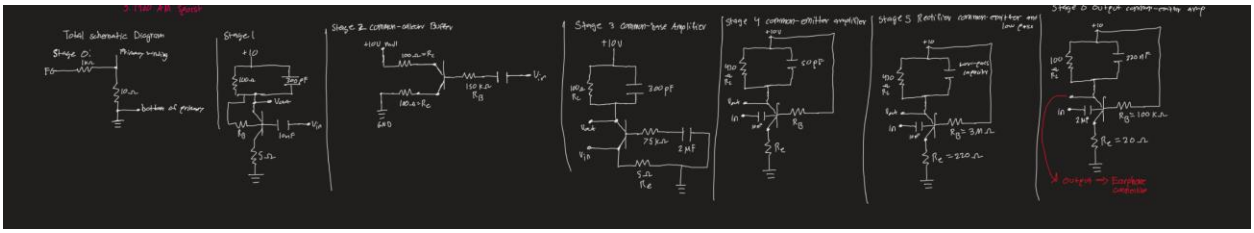


Figure 9.

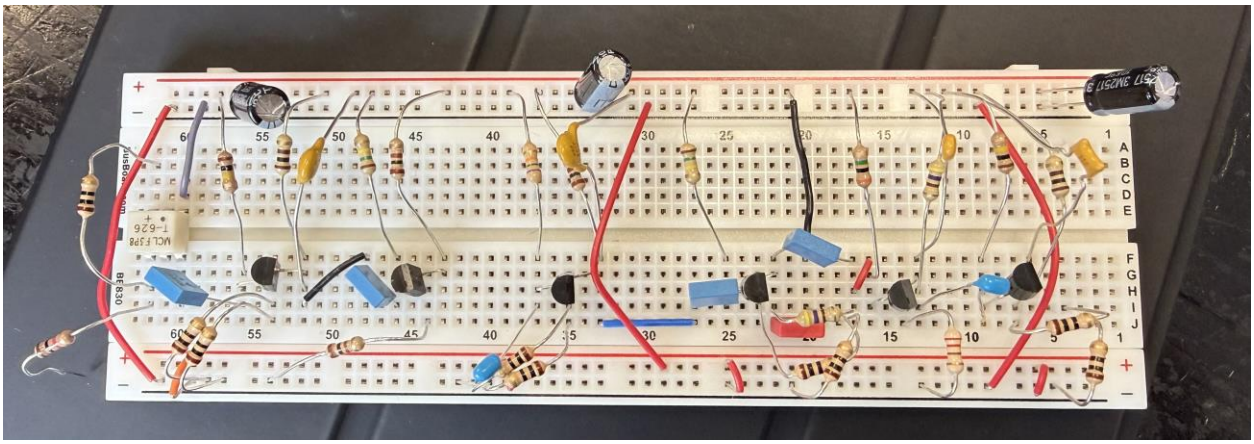


Figure 10.